

Module 5: Higher-Order Linear ODEs (Variable Coefficients)

Welcome to **Module 5**. Up until this point, we have focused exclusively on higher-order linear differential equations where the coefficients were constants (e.g., $3y'' + 2y' + y = 0$).

However, in many advanced engineering and physics problems, the coefficients are variables—specifically, powers of x . In this module, we will learn how to handle a very specific and famous type of variable-coefficient equation: the **Cauchy-Euler Equation**.

Module Learning Objectives

By the end of this module, you will be able to: 1. Identify a Cauchy-Euler Differential Equation by checking if the power of the independent variable x matches the order of the derivative it multiplies. 2. Apply the independent variable substitution ($x = e^t$) to transform the variable-coefficient equation into a standard constant-coefficient equation. 3. Solve the transformed equation using the techniques mastered in Module 4, and translate the final answer back into terms of x .

Topic Index

Navigate to the lesson for this module using the link below:

1. Cauchy-Euler Differential Equations

- **The Concept:** Recognizing the $x^n \frac{d^n y}{dx^n}$ structure.
 - **The Substitution Rule:** Letting $x = e^t$ and defining the new operator $D = \frac{d}{dt}$.
 - **The Transformation:** Changing $x \frac{dy}{dx}$ to Dy and $x^2 \frac{d^2 y}{dx^2}$ to $D(D-1)y$.
 - *Includes step-by-step solved questions converting back and forth between x and t domains.*
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Next Steps: Proceed to [Topic 1: Cauchy-Euler Equations](#) to learn how to master this transformative technique.